

ANTONIO RAPUANO

Rome, Italy • antonio.rapuano99@gmail.com • ntonioa.github.io

SUMMARY

I'm an engineer driven by curiosity and a strong desire to turn theory into practical systems. I enjoy working across the entire development pipeline – from modeling and simulation to implementation and real-world testing. I'm continuously seeking opportunities to learn, collaborate, and contribute to impactful technologies, especially in robotics and autonomous systems.

EXPERIENCE

(Details and media at ntonioa.github.io/about)

Research Engineer (Research Fellow)

Summer 2025 - Present

Sapienza University of Rome (uniroma1.it)

Highlights: Design and implementation of novel aerial interaction control strategies • Development of tactile sensing models using machine learning • Experimental validation on a hexarotor platform

Autopilot Control Systems Engineer

Fall 2024 - Present

Sapienza Flight Team (sasa-aerospace.it/flight-team)

Highlights: Design and implementation of path planning and hybrid control algorithms • Development of state estimation and filtering modules • Experimental testing on VTOL prototype

Race Timing Specialist and Team Lead

Fall 2018 - Present

ICRON (icron.it)

Highlights: Coordination of race logistics and technical setup • Supervision of start/finish line operations • Management of ranking and award procedures

EDUCATION

(Details and media at ntonioa.github.io/about)

Ph.D. in Automatic Control, Bioengineering and Operations Research

Fall 2025 - Present

Sapienza University of Rome, Rome, Italy

M.Sc. in Control Engineering

Fall 2022 - Summer 2025

Sapienza University of Rome, Rome, Italy

Final grade: 110/110 with honors

Thesis: Nonlinear predictive control of the continuum and hybrid dynamics of a suspended deformable cable for aerial pick and place (Supervisor: Prof. Antonio Franchi)

Relevant coursework: Analysis, modeling and control of nonlinear, hybrid, multivariable, networked systems • Basic and advanced robotics • Estimation • Optimal control • Machine learning • Robust control

B.Sc. in Electronic Engineering for Automation and Telecommunications

Fall 2018 - Winter 2022

University of Sannio, Benevento, Italy

Final grade: 110/110 with honors

Thesis: Implementation on FPGA of an electronic circuit controlling a mechanical arm (Supervisor: Prof. Marco Pisco)

Relevant coursework: Analog and digital electronics • Analysis, modeling and control of linear systems • Digital design • Electronic measurements • Signal processing • Telecommunications

SKILLS

Software and tools: Arduino • Blender • Fusion 360 • LTSpice • Office • Quartus Prime • Simulink • Stateflow

Programming and formatting: C/C++ • Java • HTML/CSS • LaTeX • MATLAB • Python • Verilog

Hardware: 3D printing • Circuit board soldering • Measuring

Personal skills: Attention to detail • Autonomy and initiative • Critical thinking • Curiosity and continuous learning • Problem-solving attitude • Project planning • Technical writing • Time management

LANGUAGES

Italian, native

English, fluent

Cambridge English C1 Advanced - Score 207 (Grade A, CEFR Level C2)

Summer 2022